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REPORT OF MEETINGS OF SCIENTIFIC ADVISORY PANEL
ON UNIDENTIFIED FLYING OBJECTS
CONVENED BY OFFICE OF SCIENTIFIC INTELLIGENCE, CIA
January 14 - 18, 1953

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INDEX

	<u>Page</u>
Purpose	1
Part I: History of Meetings of Panel	1
Part II: Comments and Suggestions of Panel	7
General	7
On Lack of Danger	8
Air Force Reporting System	9
Artifacts of Extraterrestrial Origin	10
Tremonton, Utah, Sighting	11
Potential Related Dangers	15
Geographic Locations of Unexplained Sightings	15
Instrumentation to Obtain Data	16
Radar Problem of Mutual Interference	18
Unexplained Cosmic Ray Phenomena	19
Educational Program	19
Unofficial Investigating Groups	23
Increase in Number of Sightings	24
Report of Panel	Tab A
List of Personnel Concerned with Meetings	Tab B
List of Documentary Evidence Presented	Tab C

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16 February 1953

MEMORANDUM FOR: Assistant Director for Scientific Intelligence
FROM : [REDACTED] 25X1A
SUBJECT : Report of Meetings of the Office of Scientific
Intelligence Scientific Advisory Panel on
Unidentified Flying Objects, January 14 - 18, 1953

PURPOSE

The purpose of this memorandum is to present:

- a. A brief history of the meetings of the O/SI Advisory Panel
On Unidentified Flying Objects (Part I),
- b. An unofficial supplement to the official Panel Report to
AD/SI setting forth comments and suggestions of the Panel
Members which they believed were inappropriate for inclusion
in the formal report (Part II).

PART I: HISTORY OF MEETINGS

GENERAL

After consideration of the subject of "unidentified flying
objects" at the 4 December meeting of the Intelligence Advisory
Committee, the following action was agreed:

"The Director of Central Intelligence will:

- a. Enlist the services of selected scientists to
review and appraise the available evidence in the
light of pertinent scientific theories...."

Following the delegation of this action to the Assistant
Director for Scientific Intelligence and preliminary investigation,

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an Advisory Panel of selected scientists was assembled. In cooperation with the Air Technical Intelligence Center, case histories of reported sightings and related material were made available for their study and consideration.

Present at the initial meeting (0930 Wednesday, 14 January)

were:

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(W&E Division), and the writer. Panel Member, [REDACTED]

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was absent until Friday afternoon. [REDACTED]

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[REDACTED] were present throughout the sessions to familiarize themselves with the subject, represent the substantive interest of their Divisions, and assist in administrative support of the meetings. (A list of personnel concerned with the meetings is given in Tab A).

WEDNESDAY MORNING

The AD/SI opened the meeting, reviewing CIA interest in the subject and action taken. This review included the mention of the O/SI Study Group of August 1952 [REDACTED]

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culminating in the briefing of the DCI, the ATIC November 21 briefing, 4 December IAC consideration, visit to ATIC (Chadwell,

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[REDACTED], and O/SI concern over potential dangers to national security indirectly related to these sightings. Mr.

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[REDACTED] enumerated these potential dangers: Following this introduction, Dr. Chadwell turned the meeting over to [REDACTED]

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as Chairman of the Panel. [REDACTED] enumerated the evidence available and requested consideration of specific reports and letters be taken by certain individuals present (Tab B). For example, case histories involving radar or radar and visual sightings were selected for [REDACTED] while reports of Green

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25X1A investigations were routed to [REDACTED] Following these remarks, the motion pictures of the sightings at Tremonton, Utah (2 July 1952) and Great Falls, Montana (15 August 1950) were shown. The meeting adjourned at 1200.

WEDNESDAY AFTERNOON

The second meeting of the Panel opened at 1400. Lt. R. S. Neasham, USN, and Mr. Harry Woo of the USN Photo Interpretation Laboratory, Anacostia, presented the results of their analyses of the films mentioned above. This analysis evolved considerable discussion as elaborated upon below. Besides Panel members and CIA personnel, Capt. E. J. Ruppelt, Dr. J. Allen Nynak, [REDACTED]

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25X1A [REDACTED] Capt. Harry B. Smith (2-e-2), and Dr. Stephen Possony were present.

Following the Photo Interpretation Lab presentation, Mr. E. J. Ruppelt spoke for about 40 minutes on ATIC methods of handling and evaluating reports of sighting and their efforts to improve the quality of reports. The meeting was adjourned at 1715.

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THURSDAY MORNING

The third and fourth meetings of the Panel were held Thursday, 15 January, commencing at 0900 with a two-hour break for luncheon. Besides Panel members and CIA personnel, Mr. Ruppelt and Dr. Hynek were present for both sessions. In the morning, Mr. Ruppelt continued his briefing on ATIC collection and analysis procedures. The Project STORK support at Battelle Memorial Institute, Columbus, was described by Dr. Hynek. A number of case histories were discussed in detail and a motion picture film of seagulls was shown. A two hour break for lunch was taken at 1200.

THURSDAY AFTERNOON

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At 1400 hours [REDACTED] gave a 40-minute briefing of Project TWINKLE, the investigatory project conducted by the Air Force Meteorological Research Center at Cambridge, Mass. In this briefing he pointed out the many problems of setting up and manning 24-hour instrumentation watches of patrol cameras searching for sightings of U.F.O.'s.

At 1615 Brig. Gen. William M. Garland joined the meeting with AD/SI. General Garland expressed his support of the Panel's efforts and stated three personal opinions:

- a. That greater use of Air Force intelligence officers in the field (for follow-up investigation) appeared desirable, but that they required thorough briefing.

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- b. That vigorous effort should be made to declassify as many of the reports as possible.
- c. That some increase in the ATIC section devoted to U.F.O. analysis was indicated.

This meeting was adjourned at 1700.

FRIDAY MORNING

The fifth session of the Panel convened at 0900 with the same personnel present as enumerated for Thursday (with the exception of Brig. Gen. Garland).

From 0900 - 1000 there was general discussion and study of reference material. Also, Dr. Hynek read a prepared paper making certain observations and conclusions. At 1000 [REDACTED] gave a briefing on his fifteen months experience in Washington as Project Office for U.F.O.'s and his personal conclusions. There was considerable discussion of individual case histories of sightings to which he referred. Following [REDACTED] presentation, a number of additional case histories were examined and discussed with Messrs. [REDACTED] Ruppelt, and Hynek. The meeting adjourned at 1200 for luncheon.

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FRIDAY AFTERNOON

This session opened at 1400. Besides Panel members and CIA personnel, Dr. Hynek was present. [REDACTED] as Panel Member, was present at this meeting for the first time. Progress of the meetings was reviewed by the Panel Chairman and tentative

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conclusions reached. A general discussion followed and tentative recommendations considered. It was agreed that the Chairman should draft a report of the Panel to AD/SI that evening for review by the Panel the next morning. The meeting adjourned at 1715.

SATURDAY MORNING

At 0945 the Chairman opened the seventh session and submitted a rough draft of the Panel Report to the members. This draft had been reviewed and approved earlier by [REDACTED] The next two and one-half hours were consumed in discussion and revision of the draft. At 1100 the AD/SI joined the meeting and reported that he had shown and discussed a copy of the initial rough draft to the Director of Intelligence, USAF, whose reaction was favorable. At 1200 the meeting was adjourned.

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SATURDAY AFTERNOON

At 1400 the eighth and final meeting of the Panel was opened. Discussions and rewording of certain sentences of the Report occupied the first hour. (A copy of the final report is appended as Tab C.) This was followed by a review of work accomplished by the Panel, and restatement of individual Panel Member's opinions and suggestions on details that were felt inappropriate for inclusion in the formal report. It was agreed that the writer would incorporate these comments in an internal report to the AD/SI. The material below represents this information.

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PART VI: COMMENTS AND SUGGESTIONS OF PANEL

GENERAL

The Panel Members were impressed (as have been others, including O/SI personnel) in the lack of sound data in the great majority of case histories; also, in the lack of speedy follow-up due primarily to the modest size and limited facilities of the ATIC section concerned. Among the case histories of significant sightings discussed in detail were the following:

Bellefontaine, Ohio (1 August 1952); Tremonton, Utah (2 July 1952); Great Falls, Montana (15 August 1950); Yaak, Montana (1 September 1952); Washington, D.C. area (19 July 1952); and Haneda A.F.B., Japan (5 August 1952), Fort Huron, Michigan (29 July 1952); and Presque Isle, Maine (10 October 1952).

After review and discussion of these cases (and about 15 others, in less detail), the Panel concluded that reasonable explanations could be suggested for most sightings and "by deduction and scientific method it could be induced (given additional data) that other cases might be explained in a similar manner." The Panel pointed out that because of the brevity of some sightings (e.g. 2-3 seconds) and the inability of the witnesses to express themselves clearly (sometimes) that conclusive explanations could not be expected for every case reported. Furthermore, it was considered that, normally, it would be a great waste of effort to try to solve most of the sightings, unless such action would benefit a training and educational program (see below). The writings of Charles Fort were referenced

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to show that "strange things in the sky" had been recorded for hundreds of years. It appeared obvious that there was no single explanation for a majority of the things seen. The presence of radar and astronomical specialists on the Panel proved of value at once in their confident recognition of phenomena related to their fields. It was apparent that specialists in such additional fields as psychology, meteorology, aerodynamics, ornithology and military air operations would extend the ability of the Panel to recognize many more categories of little-known phenomena.

ON LACK OF DANGER

The Panel concluded unanimously that there was no evidence of a direct threat to national security in the objects sighted. Instances of "Foo Fighters" were cited. These were unexplained phenomena sighted by aircraft pilots during World War II in both European and Far East theaters of operation wherein "balls of light" would fly near or with the aircraft and maneuver rapidly. They were believed to be electrostatic (similar to St. Elmo's fire) or electromagnetic phenomena or possibly light reflections from ice crystals in the air, but their exact cause or nature was never defined. Both

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[REDACTED] had been concerned in the investigation of these phenomena, but David T. Griggs (Professor of Geophysics at the University of California at Los Angeles) is believed to have been the most knowledgeable person on this subject. If the term "flying saucers" had been popular in 1943 - 1945, these objects would

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have been so labeled. It was interesting that in at least two cases reviewed that the object sighted was categorized by [REDACTED] 25X1A

25X1A [REDACTED] as probably "Foo Fighters" to date unexplained but not dangerous; they were not happy thus to dismiss the sightings by calling them names. It was their feeling that these phenomena are not beyond the domain of present knowledge of physical science, however.

AIR FORCE REPORTING SYSTEM

It was the Panel's opinion that some of the Air Force concern over U.F.O.'s (notwithstanding Air Defense Command anxiety over fast radar tracks) was probably caused by public pressure. The result today is that the Air Force has instituted a fine channel for receiving reports of nearly anything anyone sees in the sky and fails to understand. This has been particularly encouraged in popular articles on this and other subjects, such as space travel and science fiction. The result is the mass receipt of low-grade reports which tend to overload channels of communication with material quite irrelevant to hostile objects that might some day appear. The Panel agreed generally that this mass of poor-quality reports containing little, if any, scientific data was of no value. Quite the opposite, it was possibly dangerous in having a military service foster public concern in "nocturnal meandering lights." The implication being, since the interested agency was military, that these objects were or might be potential direct threats to national security. Accordingly, the need for deemphasization made itself apparent. Comments on a possible educational program are enumerated below:

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It was the opinion of [REDACTED] that the "saucer" problem had been found to be different in nature from the detection and investigation of German V-1 and V-2 guided missiles prior to their operational use in World War II. In this 1943-1944 intelligence operation (CROSSBOW), there was excellent intelligence and by June 1944 there was material evidence of the existence of "hardware" obtained from crashed vehicles [REDACTED] This evidence gave the investigating team a basis upon which to operate. The absence of any "hardware" resulting from unexplained U.F.O. sightings lends a "will-of-the-wisp" nature to the ATIC problem. The results of their investigation, to date, strongly indicate that no evidence of hostile act or danger exists. Furthermore, the current reporting system would have little value in the case of detection of enemy attack by conventional aircraft or guided missiles; under such conditions "hardware" would be available almost at once.

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ARTIFACTS OF EXTRATERRESTIAL ORIGIN

It was interesting to note that none of the members of the Panel were loath to accept that this earth might be visited by extra-terrestrial intelligent beings of some sort, some day. What they did not find was any evidence that related the objects sighted to space travelers. [REDACTED] in his presentation, showed how he had eliminated each of the known and probable causes of sightings leaving him "extra-terrestrial" as the only one remaining in many cases. [REDACTED] background as an aeronautical engineer and technical intelligence

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officer (Project Officer, BLUEBOOK for 15 months) could not be slighted. However, the Panel could not accept any of the cases sighted by him because they were raw, unevaluated reports.

Terrestrial explanations of the sightings were suggested in some cases and in others the time of sighting was so short as to cause suspicion of visual impressions. It was noted by [REDACTED] and others that extraterrestrial artifacts, if they did exist, are no cause for alarm; rather, they are in the realm of natural phenomena subject to scientific study, just as cosmic rays were at the time of their discovery 20 to 30 years ago. This was an attitude in which [REDACTED] did not concur, as he felt that such artifacts would be of immediate and great concern not only to the U. S. but to all countries. (Nothing like a common threat to unite peoples!)

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[REDACTED] noted that present astronomical knowledge of the solar system makes the existence of intelligent beings (as we know the term) elsewhere than on the earth extremely unlikely, and the concentration of their attention by any controllable means confined to any one continent of the earth quite preposterous.

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TREMONTION, UTAH, SIGHTING

This case was considered significant because of the excellent documentary evidence in the form of Kodachrome motion picture films (about 1600 frames). The Panel studied these films, the case history, ATIC's interpretation, and received a briefing by representatives of the USN Photo Interpretation Laboratory on their analysis of the film. This term had expended (at Air Force request) approximately

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1000 man-hours of professional and sub-professional time in the preparation of graph plots of individual frames of the film, showing apparent and relative motion of objects and variation in their light intensity. It was the opinion of the P.I.L. representatives that the objects sighted were not birds, balloons or aircraft, were "not reflections because there was no blinking while passing through 60° of arc" and were, therefore, "self-luminous". Plots of motion and variation in light intensity of the objects were displayed. While the Panel Members were impressed by the evident enthusiasm, industry and extent of effort of the P.I.L. team, they could not accept the conclusions reached. Some of the reasons for this were as follows:

- a. A semi-spherical object can readily produce a reflection of sunlight without "blinking" through 60° of arc travel.
- b. Although no data was available on the "albedo" of birds or polyethylene balloons in bright sunlight, the apparent motions, sizes and brightnesses of the objects were considered strongly to suggest birds, particularly after the Panel viewed a short film showing high reflectivity of seagulls in bright sunlight.
- c. P.I.L. description of the objects sighted as "circular, bluish-white" in color would be expected in cases of spectacular reflections of sunlight from convex surfaces where the brilliance of the reflection would obscure other portions of the object.

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- d. Objects in the Great Falls case were believed to have probably been aircraft, and the bright lights such reflections.
- e. There was no valid reason for the attempt to relate the objects in the Tremonton sighting to those in the Great Falls sighting. This may have been due to misunderstanding in their directive. The objects in the Great Falls sighting are strongly suspected of being reflections of aircraft known to have been in the area.
- f. The intensity change in the Tremonton lights was too great for acceptance of the P.I.L. hypothesis that the apparent motion and changing intensity of the lights indicated extremely high speed in small orbital paths.
- g. Apparent lack of guidance of investigators by those familiar with U.F.O. reports and explanations.
- h. Analysis of light intensity of objects made from duplicate rather than original film. The original film was noted to have a much lighter background (affecting relative brightness of object) and the objects appeared much less bright.
- i. Method of obtaining data of light intensity appeared faulty because of unsuitability of equipment and questionable assumptions in making averages of readings.
- j. No data had been obtained on the sensitivity of Kodachrome film to light of various intensities using the same camera type at the same lens openings.

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k. Hand "jitter" frequencies (obtainable from early part of Tremonton film) were not removed from the plots of the "single pass plots" at the end of the film.

The Panel believed strongly that the data available on this sighting was sufficient for positive identification if further data is obtained by photographing polyethylene "pillow" balloons released near the site under similar weather conditions, checking bird flight and reflection characteristics with competent ornithologists and calculating apparent "G" forces acting upon objects from their apparent tracks. It was concluded that the results of such tests would probably lead to creditable explanations of value in an educational or training program. However, the Panel noted that the cost in technical manpower effort required to follow up and explain every one of the thousand or more reports received through channels each year (1,900 in 1952) could not be justified. It was felt that there will always be sightings, for which complete data is lacking, that can only be explained with disproportionate effort and with a long time delay, if at all. The long delay in explaining a sighting tends to eliminate any intelligence value. The educational or training program should have as a major purpose the elimination of popular feeling that every sighting, no matter how poor the data, must be explained in detail. Attention should be directed to the requirement among scientists that a new phenomena, to be accepted, must be completely and convincingly documented. In other words, the burden of proof is on the sighter, not the explainer.

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POTENTIAL RELATED DANGERS

The Panel Members were in agreement with O/SI opinion that, although evidence of any direct threat from these sightings was wholly lacking, related dangers might well exist resulting from:

- a. Misidentification of actual enemy artifacts by defense personnel.
- b. Overloading of emergency reporting channels with "false" information ("noise to signal ratio" analogy-- [REDACTED])
- c. Subjectivity of public to mass hysteria and greater vulnerability to possible enemy psychological warfare.

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Although not the concern of CIA, the first two of these problems may seriously affect the Air Defense intelligence system, and should be studied by experts, possibly under ADC. If U.F.O.'s become discredited in a reaction to the "flying saucer" scare, or if reporting channels are saturated with false and poorly documented reports, our capability of detecting hostile activity will be reduced.

Dr. Page noted that more competent screening or filtering of reported sightings at or near the source is required, and that this can best be accomplished by an educational program.

GEOGRAPHIC LOCATIONS OF UNEXPLAINED SIGHTINGS

The map prepared by ATIC showing geographic locations of officially reported unexplained sightings (1952 only) was examined by the Panel. This map showed clusters in certain strategic areas such as Los Alamos. This might be explained on the basis of 24-hour watchful guard and

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awareness of security measures near such locations. On the other hand, there had been no sightings in the vicinity of sensitive related AE establishments while there were occasionally multiple cases of unexplained sightings in non-strategic areas. Furthermore, there appeared to be no logical relationship to population centers. The Panel could find no ready explanation for these clusters. It was noted, however, that if terrestrial artifacts were to be observed it would be likely that they would be seen first near foreign areas rather than central U. S.

INSTRUMENTATION TO OBTAIN DATA

The Panel was of the opinion that the present AFIC program to place 100 inexpensive 35 mm. stereo cameras in the hands of various airport control tower operators would probably produce little valuable data related to U.F.O.'s. However, it was recognized that such action would tend to allay public concern in the subject until an educational program had taken effect. It was believed that procurement of these cameras was partly the result of public pressure in July 1952. With the poor results of the year-long Project TWINKLE program of 24-hour instrumentation watch (two frames of film showing nothing distinguishable), a widespread program of sky-watching would not be expected to yield much direct data of value.

There was considerable discussion of a possible "sky patrol" by amateur astronomers (Hynek) and by wide-angle cameras [REDACTED] and [REDACTED] pointed out that at present a considerable fraction

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of the sky is now--and has been for many years--under surveillance every clear night in several meteor and aurora observing programs as well as sky mapping programs at the various locations listed below. Although the attention of those astronomers is largely directed toward identified rather than unidentified objects, no case of any striking unidentified object is known to [REDACTED] or Dr. Hynek. Such an object would most certainly be reported if found on patrol plates.

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A case was cited where an astronomer refused to interrupt his exposure in order to photograph an alleged sighting in a different part of the sky. This led Dr. Hynek to say that, if a program of watching could be an adjunct of planned astronomical programs, little cost would be involved and that the trained astronomical personnel might photograph a sighting of an unidentified object.

The location of some of these programs and their directors are believed to be:

- a. Harvard University, Cambridge and New Mexico (meteor patrol) - Whipple.
- b. Yerkes Observatory, University of Chicago and Fort Davis, Texas (several programs)--Meinel (aurorae), Kuiper (asteroids), Morgan (wide angle camera).
- c. University of Alaska, Fairbanks (aurorae) - Elvey
- d. Dominion Observatory, Ottawa (meteors) - Millman
- e. Palomar Observatory, California (sky map) - Minkowski
- f. Lick Observatory, California (sky map) - Shane

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It was agreed by the Panel that no government-sponsored program of optical nation-wide sky patrol is worthwhile at the present time, and that the encouragement of amateur astronomers to undertake such a program might have the adverse effect of over-emphasizing "flying saucer" stories in the public mind. However, the issue of radar'scope cameras for recording peculiar radar echoes would serve several purposes, including the better understanding of radar interference as well as identification of U.F.O.'s.

RADAR PROBLEM OF MUTUAL INTERFERENCE

25X1A This characteristic problem of radar operation wherein the pulse signal (of approximately the same frequency) from station A may be picked up on the screen of station B and show as a high-speed track or series of dots was recognized to have probably caused a number of U.F.O. reports. This problem was underlined by information received indicating ADC concern in solving this problem of signal identification before service use of very high-speed aircraft or guided missiles (1955-1956). [REDACTED] believed that one answer to this problem 25X1A was the use of a "doppler filter" in the receiving circuit. [REDACTED] suggested that the problem might be better solved by the use of a "controlled jitter" wherein the operator receiving "very fast tracks" (on the order of 1000 - 10,000 m.p.h.) would operate a circuit which would alter slightly his station's pulse frequency rate. If the signal received on the screen had been caused by mutual interference with another station, the track would now show itself at a different distance

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from the center of the screen, if it still appeared at all. [REDACTED] felt such a technical solution was simpler and would cost much less than a "Doppler filter."

UNEXPLAINED COSMIC RAY PHENOMENA

Two reported cases were examined: one at Palomar Mountain, California, in October 1949, when cosmic ray counter went "off scale for a few seconds," apparently while a "V" of flying saucers was observed visually; and two, a series of observations by the "Los Alamos Bird Watchers Association" from August 1950 to January 1951, when cosmic ray coincidence counters behaved queerly. Circuit diagrams and records were available for the latter, and [REDACTED] was able quickly to point out that the recorded data were undoubtedly due to instrumental effects that would have been recognized as such by more experienced observers.

The implication that radioactive effects were correlated with unidentified flying objects in these two cases was, therefore, rejected by the Panel.

EDUCATIONAL PROGRAM

The Panel's concept of a broad educational program integrating efforts of all concerned agencies was that it should have two major aims: training and "debunking".

The training aim would result in proper recognition of unusually illuminated objects (e.g., balloons, aircraft reflections) as well as natural phenomena (meteors, fireballs, mirages, noctilucent clouds). Both visual and radar recognition are concerned. There would be many

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levels in such education from enlisted personnel to command and research personnel. Relative emphasis and degree of explanation of different programs would correspond to the categories of duty (e.g., radar operators; pilots; control tower operators; Ground Observer Corps personnel; and officers and enlisted men in other categories.) This training should result in a marked reduction in reports caused by misidentification and resultant confusion.

The "debunking" aim would result in reduction in public interest in "flying saucers" which today evokes a strong psychological reaction. This education could be accomplished by mass media such television, motion pictures, and popular articles. Basis of such education would be actual case histories which had been puzzling at first but later explained. As in the case of conjuring tricks, there is much less stimulation if the "secret" is known. Such a program should tend to reduce the current gullibility of the public and consequently their susceptibility to clever hostile propaganda. The Panel noted that the general absence of Russian propaganda based on a subject with so many obvious possibilities for exploitation might indicate a possible Russian official policy.

Members of the Panel had various suggestions related to the planning of such an educational program. It was felt strongly that psychologists familiar with mass psychology should advise on the nature and extent of the program. In this connection, Dr. Hadley Cantril (Princeton University) was suggested. Cantril authored "Invasion from Mars", (a study in the

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psychology of panic, written about the famous Orson Welles radio broadcast in 1938) and has since performed advanced laboratory studies in the field of perception. The names of Don Marquis (University of Michigan) and Leo Rosten were mentioned as possibly suitable as consultant psychologists. Also, someone familiar with mass communication techniques, perhaps an advertising expert, would be helpful. Arthur Godfrey was mentioned as possibly a valuable channel of communication reaching a mass audience of certain levels.

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[REDACTED] suggested the U. S. Navy (CNR) Special Devices Center, Sands Point, L. I., as a potentially valuable organization to assist in such an educational program. The teaching techniques used by this agency for aircraft identification during the past war was cited as an example of a similar educational task. The Jam Handy Co. which made World War II training films (motion picture and slide strips) was also suggested, as well as Walt Disney, Inc. animated cartoons. Dr. Nynek suggested that the amateur astronomers in the U. S. might be a potential source of enthusiastic talent "to spread the gospel". It was believed that business clubs, high schools, colleges, and television stations would all be pleased to cooperate in the showing of documentary type motion pictures if prepared in an interesting manner. The use of true cases showing first the "mystery" and then the "explanation" would be forceful.

To plan and execute such a program, the Panel believed was no mean task. The current investigatory group at ATIC would, of necessity, have to be closely integrated for support with respect to not only the

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historical cases but the current ones. Recent cases are probably much more susceptible to explanation than older ones; first, because of ATIC's experiences and, secondly, their knowledge of most plausible explanations. The Panel believed that some expansion of the ATIC effort would certainly be required to support such a program. It was believed inappropriate to state exactly how large a Table of Organization would be required. Captain Ruppelt of ATIC unofficially proposed, for purposes of analyzing and evaluating reports:

- a. An analysts' panel of four officers
- b. Four officer investigators
- c. A briefing officer
- d. An ADC liason officer
- e. A weather and balloon data officer
- f. An astronomical consultant
- g. A Group Leader, with administrative assistant, file clerks and stenographers.

This proposal met with generally favorable comment. The Panel believed that, with ATIC's support, the educational program of "training and debunking" outlined above might be required for a minimum of one and one-half to two years. At the end of this time, the dangers related to "flying saucers" should have been greatly reduced if not eliminated. Cooperation from other military services and agencies concerned (e.g., Federal Civil Defense Administration) would be a necessity. In investigating significant cases (such as the Tremonton, Utah, sighting), controlled experiments might be required. An example

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would be the photographing of "pillow balloons" at different distances under similar weather conditions at the site.

The help of one or two psychologists and writers and a subcontractor to produce training films would be necessary in addition. The Panel considered that ATIC's efforts, temporarily expanded as necessary, could be most useful in implementing any action taken as a result of its recommendations. Experience and records in ATIC would be of value in both the public educational and service training program envisaged. Dr. Robertson at least was of the opinion that after public gullibility lessened and the service organizations, such as ADC, had been trained to sift out the more readily explained spurious sightings, there would still be a role for a very modest-sized ATIC section to cope with the residuum of items of possible scientific intelligence value. This section should concentrate on energetically following up (perhaps on the advice of qualified Air Force Scientific Advisory Board members) those cases which seemed to indicate the evidence of unconventional enemy artifacts. Reports of such artifacts would be expected to arise mainly from Western outposts in far closer proximity to the Iron Curtain than Lubbock, Texas!

UNOFFICIAL INVESTIGATING GROUPS

The Panel took cognizance of the existence of such groups as the "Civilian Flying Saucer Investigators" (Los Angeles) and the "Aerial Phenomena Research Organization (Wisconsin). It was believed that such organizations should be watched because of their potentially

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great influence on mass thinking if widespread sightings should occur. The apparent irresponsibility and the possible use of such groups for subversive purposes should be kept in mind.

INCREASE IN NUMBER OF SIGHTINGS

The consensus of the Panel was, based upon the history of the subject, that the number of sightings could be reasonably expected to increase again this summer.

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TAB A

REPORT OF THE SCIENTIFIC PANEL
ON
UNIDENTIFIED FLYING OBJECTS

Released in MORI

Jan 17 1953

1. Pursuant to the request of the Assistant Director for Scientific Intelligence, the undersigned Panel of Scientific Consultants has met to evaluate any possible threat to national security posed by Unidentified Flying Objects ("Flying Saucers"), and to make recommendations thereon. The Panel has received the evidence as presented by cognizant intelligence agencies, primarily the Air Technical Intelligence Center, and has reviewed a selection of the best documented incidents.

2. As a result of its considerations, the Panel concludes:

a. That the evidence presented on Unidentified Flying Objects shows no indication that these phenomena constitute a direct physical threat to national security.

We firmly believe that there is no residuum of cases which indicates phenomena which are attributable to foreign artifacts capable of hostile acts, and that there is no evidence that the phenomena indicate a need for the revision of current scientific concepts.

3. The Panel further concludes:

a. That the continued emphasis on the reporting of these phenomena does, in these parlous times, result in a threat to the orderly functioning of the protective organs of the body politic.

We cite as examples the clogging of channels of communication by irrelevant reports, the danger of being led by continued false alarms to ignore real

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indications of hostile action, and the cultivation of a morbid national psychology in which skillful hostile propaganda could induce hysterical behavior and harmful distrust of duly constituted authority.

4. In order most effectively to strengthen the national facilities for the timely recognition and the appropriate handling of true indications of hostile action, and to minimize the concomitant dangers alluded to above, the Panel recommends:

a. That the national security agencies take immediate steps to strip the Unidentified Flying Objects of the special status they have been given and the aura of mystery they have unfortunately acquired;

b. That the national security agencies institute policies on intelligence, training, and public education designed to prepare the material defenses and the morale of the country to recognize most promptly and to react most effectively to true indications of hostile intent or action.

We suggest that these aims may be achieved by an integrated program designed to reassure the public of the total lack of evidence of inimical forces behind the phenomena, to train personnel to recognize and reject false indications quickly and effectively, and to strengthen regular channels for the evaluation of and prompt reaction to true indications of hostile measures.

/s/ H. P. Robertson, Chairman
California Institute of Technology

/s/ Lloyd V. Berkner
Associated Universities, Inc.

/s/ Luis W. Alvarez
University of California

/s/ S. A. Goudsmit
Brookhaven National Laboratories

/s/ Thornton Page

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NEW SOVIET WEAPONS

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The following is a report of the experiences of a German scientist, Dr H. Tellman, who was taken prisoner by the Russians in 1943 and who worked for them as a scientist until he escaped by airplane. Dr Tellman now lives in Argentina.

After my capture in 1943, I was interviewed in the Chakalov prisoners' camp and offered a position as scientist in the service of the Soviets. After having been given assurances that my work would not be directly connected with the armament industry, at least for the duration of the war, I accepted.

My first job, to which I was sent 2 weeks later, was at Molotov, where a huge aluminum plant was being built. Its power supply was to come from the hydroelectric stations of the Central Urals. Work went on 24 hours a day, weather permitting, most of it being carried out by women. My task was the supervision of the laying of the high-power lines which went all the way into the Urals. The machines in the plant was nearly all American made, and had been brought in via Siberia.

At the beginning of 1944, I was requested to work for the Cosmic Ray Laboratory at Tbilisi, which was operating under the supervision of the Ministry of the Armed Forces. The institute not further identified was housed in a former palace in Tbilisi, and was directed by a well-known Russian scientist who had spent much time in Germany and spoke German well. The altitude laboratory of the institute was located in the mountains at an altitude of more than 3,000 meters. The work consisted of investigation of the nature of cosmic rays and observation of the changes in their intensity during the course of one year.

The institute was inspected by a commission of politicians and scientists from Moscow. An old colleague of mine who had studied with me at the University of Berlin before the war turned out to be a member of this commission, and offered me a position in Moscow. Six weeks later I was called to Moscow by the War Academy. At first I was busy only with conferences in the War Academy and

- 1 -

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in offices of political authorities. They seemed to be trying to get a good picture of my scientific capabilities and political opinions. I was also given indoctrination in Soviet ideology, and my friend told me confidentially that it was necessary for me to attend party functions, but he added: "You can think of whatever you like while you are there." A few days after these preliminaries I was assigned to a newly founded commission, charged with supervising development in various special fields and with operating as a consulting authority in the case of difficulties. The commission consisting of 14 scientists, nearly all of them trained abroad, either in Germany or in the US, who had been working in these countries for very long periods of time, and who were all very capable specialists.

My friend, Professor D. took me along on an inspection trip to Tomsk where the Russians had built an experimental station for V-2 rockets. I met an old acquaintance from Germany, Engineer P., who was in charge of the technical work of the entire installation. At first, the only experiments performed were the same kind which I knew from my work in Germany, but later, during my stay, new designs with much higher power were developed. Work was also carried out in the field of radio guidance of rockets. I saw the launching of several giant rockets whose dimensions considerably exceeded those of the V-2. The measuring devices recorded ceiling altitudes of 210 kilometers in these flights. A new rocket is still in the development stage. It weighs 40 tons and is expected to reach an altitude of more than 400 kilometers. I was also interested in the design of a multistage rocket, whose first stage was to be powered by nuclear energy.

To study the effect of the high accelerations on the living organism, the Russians placed into the warhead a parrot in an insulated cage. The warhead otherwise housed the recording instruments. The parrot was found to have suffered no ill effects.

The experimental rockets in the development stage today are nearly powerful enough to fly from the earth to the moon and to leave the earth's gravitational field.

A second trip took me to Kalinin. A large rocket-aircraft-testing station has been built there. The rocket aircraft developed there are based on the design of the German V-1. Essentially, they are nothing but manned rockets. The wings are small and sweptback. The cockpit is hermetically sealed and holds a two-man crew. The Soviets have also succeeded in obtaining the services of most of the important German rocket experts, who have attained very high speeds with their rocket aircraft; in one case, an aircraft equipped with three rocket engines reached a speed of nearly 2,000 kilometers per hour.

During one test flight which I witnessed, the material did not withstand the terrific strain, and the aircraft crashed, killing the German pilot. These aircraft are started from catapults. Fuel consists of a hydrocarbon compound and nitric acid. One of the German test pilots succeeded in reaching an altitude of 25,000 meters with one of these aircraft. The immense power of the engines was demonstrated to me when a rocket aircraft turned over during take-off and exploded. The explosion made a crater of 15 meters in diameter and 3 meters in depth, and the aircraft was torn into tiny fragments. These rocket aircraft are still in the experimental stage. However, the USSR has many types of jet aircraft which are already in service. They were designed and built in a plant near Voronezh by former members of the Messerschmitt firm.

One of my inspection trips took me back to the cosmic radiation institute at Tbilisi. My former chief there told me that he had lost both his sons as a result of bacteria experiments. One of them was a bacteriologist, the other a doctor. They had been ordered to set up a laboratory on a small island near Baku where dangerous bacteria were to be investigated and cultures grown. One

- 2 -

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of their colleagues infected himself with a culture, and transmitted the disease to the rest of the staff, so that all 40 members of the staff died within a few hours after terrible agony. The laboratory, I was told, now has a new staff, but my friend did not know what work was being conducted there.

With Professor D., I witnessed a test of a new weapon in the Caucasus. We went to the edge of a deep valley, which had vertical walls almost 300 meters high. Heavy smoke was generated in the valley by means of rocket projectiles fired from an adjacent plateau. Then another type of rocket was fired into the smoke. These second rockets generated a dense white vapor. As soon as the vapor mixed with the smoke, a terrific thunderstorm broke loose below us. We had to wear special gas masks. Lightning and thunder followed in uninterrupted succession, and the entire area seemed to be a mass of flame. Whenever the storm let up, new rockets were fired and its intensity increased again. The storm lasted for 2 hours.

When we investigated the area the next day, we found that not one rock, not one tree or shrub had been spared by the lightning bolts. Nobody could have survived in that hell. It was obvious that the electric charges had always gone from the cloud to the ground, a proof that the cloud must have been electrically charged. These artificial thunderstorms are designed to take the place of artillery bombardment in mountainous areas, where, as was shown during World War II, the effectiveness of artillery is limited, because the terrain offers too much opportunity for cover. However, there is no cover against this new weapon.

During a conversation with Professor D., I stated once that I would like to get out of the "golden cage" and rejoin my family in Germany. He answered that it would be better if I had my family come to Moscow instead, since he had it on good authority that none of the German scientists and workers in the Soviet Union would be allowed to leave the country, because they knew too much and the Soviets would not risk letting this information out of the country. When I answered that I would simply escape from the country if I was not allowed to leave legally, he warned me that the controls were very strict and that the frontiers had recently been heavily fortified. He further said that I would only be imprisoned or perhaps executed for my attempt.

The day after this conversation, D. and I went to Novorossiysk to attend a special experiment. We went aboard a small cruiser. We could get no definite information on the nature of the experiment, but the presence of a large number of high staff officers and political functionaries indicated that it was a test of special importance. After we had put out to sea, we were met by a flotilla of six large submarines which took us on a southerly course. After we had lost sight of the coastline, the convoy spread out. The submarines submerged after traveling another mile, surfaced again after half an hour, returned to the cruiser to make a report, and then left for the north. Suddenly, about half a mile in front of us, a steep wall of water rose from the calm sea, and moved rapidly toward the south. The tidal wave was about 20 meters high, and would have wrecked any vessel within its path. The weird part of the demonstration was the fact that the wave did not expand, either laterally or to the rear, and rose up out of the water like a solid wall without disintegrating, in contrast to the phenomenon observed in underwater explosions where the water rises up in a fountain. It was also peculiar that we noticed no concussion and no explosion, and that we felt only a slight swell on our cruiser.

I tried to obtain some information on the experiment, but the answers were evasive, and Professor D. finally put an end to my questions by saying: "It is not good to want to know everything right away. In due time you will realize how important it was for you to attend this test." I gathered from the conversations among other observers that this had been a test of a means of warfare and that great results were expected from its further development.

- 3 -

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In the USSR, submarine construction, together with aircraft construction further development of the V-2 rocket, and the development of a practicable atomic bomb, is a priority project of the armament industry. Prior to the war, the Russian built, at Leningrad, only the standard type submarine of 900 tons displacement, not counting a few experimental types. Only after the war, when the Russians had occupied Berlin and the German Baltic ports and had obtained the German design data and ships, did they take up submarine construction intensively. In 1946, the Russians succeeded in getting a number of the members of the staff of the Glueckauf Engineering Office at Blankenburg in the Harz Mountains -- during the war the central designing office for German submarines -- to return there by offering them high salaries. The construction of the German Type 26 submarine, the so-called Walther submarine, was then resumed. Some of the Blankenburg staff later went to Koenigsberg and to Leningrad, others returned to the Western zone.

The captured U-boats were concentrated at Leningrad and at Kronshtadt. The Russians are working on further development of the German Type 21 submarine. In the summer of 1948, about 15 submarines of this type were under construction at the Leningrad Navy Yard. An interesting improvement instituted by the Russians is the quick-loading mechanism of the bow torpedo tubes. The German quick-loading mechanism did not function properly, but the Russians have improved it to such an extent that all six tubes in the box can now fire three times within 3 minutes.

The Russians are also experimenting with the installing of rocket launchers on Type 21 submarines. The launcher resembles a torpedo tube. It is set up in front of the conning tower, and operated from inside the ship. It will still function when the ship is submerged to half the height of the conning tower. The rockets are powered by liquid fuel, consisting of a hydrocarbon -- nitric acid mixture with automatic ignition. Fuel feed is by compressed air. The rockets have a range of about 7 kilometers, and their accuracy is adequate.

At Odessa, the Russians are building two-man submarines developed from the German design. They are powered by a high-rpm diesel engine. The captain of the submarine can stand up on his seat when the ship is surfaced, and the upper part of his body protrudes from the conning tower. The engineer sits in front of the engine, and can start the engine with a hand crank, in case the battery is run down and no longer has enough power to operate the self-starter. The tower is equipped with a short periscope which is retractable. The submarine carries two torpedoes underneath the hull. This type of submarine is manned only by volunteers. It is capable of a speed of about 12 nautical miles.

The Russians are testing three new types of torpedoes. The first is the well-known acoustic torpedo, which is automatically attracted to the target by the noise of the target's propellers. This version has been developed to a high degree of perfection, and possesses high accuracy. Of course, it still suffers from the disadvantage that the target can tow noise-producing dummies which will deflect the torpedo, provided they operate in the proper frequency range.

The second version used ultrasonic guide beams. The frequency is in the range of 800 kc. The results obtained with this torpedo were very satisfactory up to a distance of 3 nautical miles. The torpedo is not very susceptible to jamming, unless the jamming is tuned exactly to the guide-beam frequency. At large distances, the guidance becomes less reliable, because it is difficult to produce ultrasonic waves at that frequency with a sufficiently large range.

The third version employs infrared radiation for automatic steering, and uses the principle of radar. This design was first developed by the Germans toward the end of the war and then perfected by the German engineers for the Russians. The torpedo is very accurate, even at long ranges, provided the infrared ray generator functions properly. Since this is not always the case, 35 percent of the torpedoes misfire.

- 4 -

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From the Black Sea, I was called to the Aral Sea to attend another experiment. On reaching the Aral Sea, I was fitted with arctic clothing. We set up some measuring instruments and photographic equipment on a hill overlooking the water. A squadron of bombers dropped about four 250-kilogram bombs about 2 kilometers out into the sea. A few moments after the bombs hit, the water suddenly became very calm, and seemed to be freezing over, while the normal motion of the water continued 3 or 4 kilometers further out, as we could observe through our field glasses. The waves became bigger, obviously because a strong wind had sprung up out there. The bay in front of us began to become foggy, and we went closer, because visibility was getting poor.

At a distance of about 400 meters from the shore, we were hit by a very cool flow of air, and the fog became progressively denser. The fog began to move violently, resembling the smoke over a great fire. The temperature kept dropping. While it had been +18° C before the start of the experiment, it had now dropped to -26° C at a distance of 300 meters from the shore. We froze, despite our arctic clothing. At another 100 meters closer to the shore, it had dropped to -43° C, and after only ten more steps, to -50° C. There was no point in going further, because the fog was now limiting visibility to one meter. The wind had become a gale. On the way back to the hill, we got lost in the fog, and took a long time to find our original observation point.

The fog lifted after 4 hours, and the temperature began to rise again, so that we could finally go down to the shore. The entire bay was covered by a thick sheet of ice, extending over about one square kilometer. The ice was over half a meter thick at the points of impact of the bombs. The waves created by the high wind had solidified in the middle of their motion, and the sea resembled a jagged arctic landscape. The temperature over the bay was still very low, and we were told that it would not thaw out for several days. Such bombs or rockets, when used against harbors, must have disastrous results.

A few days after my return to Moscow, I was taken to Peenemuende. The installation is operating full blast, and the region between Usedom and Greifswald is one single armed camp. There is no trace left of the demolitions carried out in 1945. Over 150 German scientists are working around the clock developing rocket projectiles and rocket-propelled fighter aircraft. I was able to determine that the Russians had obtained all German data for all the versions of the V-2 rocket. Special attention was given to the A-8 version of this rocket, which can fire 6,000 kilometer across the Atlantic with a flight time of 42 minutes.

The tests with guided rockets at Peenemuende made a great impression on me. Rockets were launched from sites in the Leningrad-Kronshtadt area. They landed with almost dead accuracy on the island of Pol. The rockets are launched to an altitude of 12,000 meters, and the propulsion unit of the rocket is cut in at that altitude by radio signal. The rockets then fly in a straight line, controlled by radio and radar signals from picket boats in the Baltic, until a measuring station stops them over the island of Pol and breaks off the flight there. The rockets come down nearly vertically and land near the target. The tests were repeated several times, and the results, in regard to accuracy, were always equally good. I am convinced now that the sensational reports once heard in Germany about rockets over the Baltic were not just imagination, but that rockets from Leningrad sometimes flew as far as Swinemuende, and that some of them supposedly got lost and flew to Sweden.

My work at Peenemuende consisted mostly in trying to persuade the German scientists there to go to work for the War Academy in Moscow. In 1945, the Russians shanghaied the scientists, but stopped this practice in 1948, since they found out that they could not obtain good work from scientists who were in the USSR under duress. During my conversations with my colleagues there I found that the research results obtained at Peenemuende should be a matter of great concern to other countries. The accuracy attained with rockets over the comparatively short range of 1,100 kilometers between Leningrad and Usedom was also attained

- 5 -

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over a 2,500-kilometer range. These tests were conducted between the Kronshadt-Leningrad area and the great rocket-testing site at Omsk in Siberia. On the basis of these tests, it could easily be possible for the Russian using rockets launched from the interior of the USSR, to reach any target in Europe or in the US with great accuracy.

I had not given up my plans for escape. I could not bear working under constant political pressure, and I wanted to spare my family from moving to USSR and did not want to expose them to an uncertain fate there.

I was helped by a coincidence. In July 1948 I was at Kharkov, where I had to inspect equipment in the huge power plant. There I met a German engineer named Wintersdorff, a former pilot, who had been shanghaied by the Russians during a visit to Soviet Zone Germany in 1946, and who was plotting an escape by air to Athens where he had a brother-in-law. By luck, a high official of the Ministry of War whom I met at Kharkov invited me to fly back to Moscow with him in his private plane, and agreed to take Wintersdorff, my "assistant" along. During the flight, with the aid of a revolver which Wintersdorff had obtained, we overpowered the official, the pilot, and the radio operator, put parachutes on them, and threw them from the plane. With Wintersdorff at the controls, we flew to Athens, from where I reached Germany.

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- 6 -

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